

LESSON 4

# **MERTON (1976) MODEL**

Derivative Pricing - Module 7



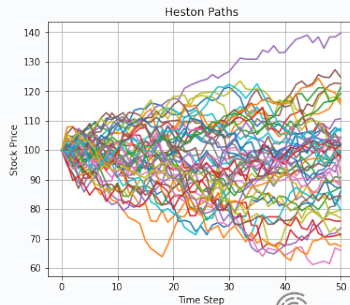
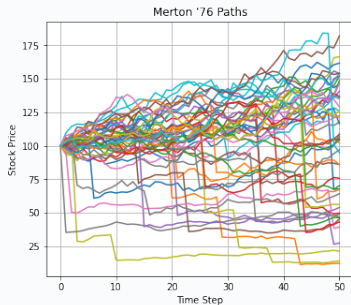
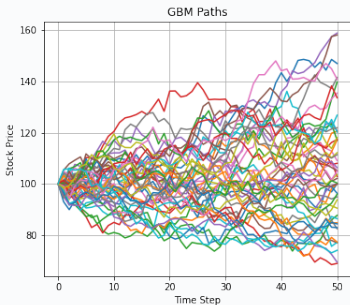
# Outline

- ▶ Implementing the Merton model in Python
- ▶ Comparison to other models
- ▶ Discussion of model parameters

# Merton Model Implementation

⇒ Go to the associated Jupyter Notebook for this lesson to explore these topics in depth.

- One of the things you'll see there is how the price paths produced by the Merton model compare to the classic GBM or Heston:



# Summary of Lesson 4

In Lesson 4, we have looked at:

- ▶ Merton jump diffusion model
- ▶ Model parameters on their own and in comparison to previous models

⇒ **TO-DO NEXT:** Go to the associated Jupyter Notebook if you haven't done so yet.

⇒ **That's all for the Derivative Pricing course!** In the next course, we will go deeper into these most recent topics by exploring other models combining Heston's stochastic volatility with Merton's jump diffusion (e.g., BCC or Bates models). Also, we will cover the important topic of learning how to properly calibrate these complex models, a worthwhile endeavor for any quant!